Module 4: Proportions and Sample-size Planning

Lesson 4 | Estimate a share and plan precision

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Learning objectives and reading

Construct a large-sample proportion interval, check its conditions, and plan a sample size for a proportion or mean. Read §§8.3–8.4, especially printed pp.378–381. The proportion interval below reproduces the book's normal approximation; it is not a universal method for small counts.

Business case: women golfers and tee times

The textbook reports a national survey of **900 women golfers**, of whom **396** were satisfied with the availability of tee times. What interval estimates the population proportion satisfied? How many completed responses would a new survey need for a margin of error of **0.025** at **95% confidence**? We use the published counts and planning question, not verified individual responses or an independently checked survey design.

يعرض الكتاب مسحًا وطنيًا شمل **900 لاعبة غولف**، كانت **396 منهن** راضيات عن توافر مواعيد بدء اللعب. ما الفترة التي تقدّر نسبة الراضيات في المجتمع؟ وكم استجابة مكتملة يحتاجها مسح جديد لهامش خطأ **0.025** عند مستوى ثقة **95**؟ نستخدم الأعداد وسؤال التخطيط المنشورين، وليس استجابات فردية جرى التحقق منها أو تصميم مسح تحقّقنا منه بصورة مستقلة.

From a binary response to a proportion

Define success before counting: here it means **satisfied**, not every favorable response on an unspecified question. For x successes among n responses,

$$\hat{p} = \frac{x}{n}, \qquad \widehat{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

$$\hat{p} \pm z_{1-\alpha/2}\widehat{SE}.$$

This is the normal or **Wald interval**. It assumes a design supporting independent binary responses or an appropriate small sampling fraction, and a meaningful target population. Check $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$ in this course. A large count does not establish representative selection or eliminate nonresponse bias.

```
# c(): combines values into a vector, converting them to a common type when needed.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# The success denominator includes all 900 surveyed women golfers, not only the satisfied
 ↪  respondents.
# qnorm(.975) is the two-sided 95% normal critical value; sqrt() computes the estimated
 ↪  SE.
x <- 396
n <- 900
phat <- x/n
se <- sqrt(phat*(1-phat)/n)
margin <- qnorm(.975)*se
ci <- phat+c(-1,1)*margin
knitr::kable(data.frame(Estimate=phat, Successes=x, Failures=n-x,
 SE=se, Margin=margin, Lower=ci[1], Upper=ci[2]),digits=5)
```

Estimate	Successes	Failures	SE	Margin	Lower	Upper
0.44	396	504	0.01655	0.03243	0.40757	0.47243

Both counts (396 and 504) are comfortably above 10. The interval is approximately **0.4076 to 0.4724**, or **40.76% to 47.24%**. The margin is about **3.24 percentage points**, not 3.24% of the estimated proportion.

If there are very few successes or failures, the Wald approximation may fail badly, including zero width or endpoints outside [0,1]. Do not silently clip endpoints and claim that coverage is repaired. Score or exact binomial methods are possible extensions. R's default `prop.test()` uses a score-based approach with a continuity correction, so its interval need not equal this textbook Wald calculation.

Plan a proportion sample size

Before collecting new data, choose confidence, desired margin E, and a planning value p^*:

$$n = \left\lceil \frac{z_{1-\alpha/2}^2 p^*(1-p^*)}{E^2} \right\rceil.$$

The ceiling rounds **up** because the minimum required size must be met. A prior survey can suggest p^* but cannot guarantee the new population share. If no defensible value is available, $.50$ maximizes $p^*(1-p^*)$ and is conservative within this simple-sampling normal approximation.

```
# qnorm(): returns a normal lower-tail quantile; sd is standard deviation, not variance.
# c(): combines values into a vector, converting them to a common type when needed.
# kable(): formats a data frame or matrix as a readable report table; digits sets
 ↪  displayed precision.
# data.frame(): builds a table whose named columns have compatible lengths.
# ceiling() rounds upward to a whole number of completed responses; E=.025 means 2.5
 ↪  percentage points.
```

```r
# Compare the prior-survey planning share with .50; exact R quantiles may differ slightly
 ↪  from rounded tables.
z <- qnorm(.975)
E <- .025
planning_p <- c(.44,.50)
required <- ceiling(z^2*planning_p*(1-planning_p)/E^2)
knitr::kable(data.frame(Planning_share=planning_p, Completed_responses=required))
```

Planning_share	Completed_responses
0.44	1515
0.50	1537

The required completed samples are **1,515** using .44 and **1,537** using .50, matching the book's recommendations. This controls the planned normal-approximation margin under the design assumptions. It does not guarantee response or remove coverage bias. Invitations must exceed completed responses if some invitees do not respond; nonresponse adjustment requires a justified response-rate assumption.

Plan precision for a mean

With a defensible planning SD σ^*, the normal-based planning formula is

$$n = \left\lceil \left(\frac{z_{1-\alpha/2}\sigma^*}{E} \right)^2 \right\rceil.$$

Use E and σ^* in the **same measurement units**. A pilot or relevant historical source may provide the planning SD. This is a planning approximation when SD is estimated; a final unknown-SD mean interval still uses t. Complex designs and substantial sampling fractions need their corresponding design calculations.

For an **original planning extension** of the checked Lloyd's example, take its $20 SD and request a $2 margin at 95% confidence:

```r
# qnorm(): returns a normal lower-tail quantile; sd is standard deviation, not variance.
# Use the checked Lloyd's SD in an explicitly new planning question, not a new observed
 ↪  sample.
# Squaring gives required n; ceiling() preserves the requested maximum planned margin.
ceiling((qnorm(.975)*20/2)^2)
```

```
[1] 385
```

This gives 385 completed observations under the stated assumptions. Halving E roughly quadruples the required sample size. More precision has a cost; select a practically useful margin before collecting data.

Choose the method after learning the conditions

Target and inputs	Interval or planning tool	Key check
Mean, known population SD	Normal z interval	SD genuinely treated as known; valid design
Mean, unknown population SD	t interval with n−1 df	Outliers, shape, independence, sampling design
Binary proportion, adequate counts	Textbook normal/Wald interval	Success/failure counts and representative design
Planned mean precision	z-based planning SD formula	Units and credibility of SD
Planned proportion precision	Planning share formula	E in proportion units; round up

Sources and limits

Anderson et al. (2020), §8.4, p.378 (396/900 and interval), pp.380–381 (E=.025 and sample sizes). Mean planning formula: §8.3; the Lloyd's $2 margin is an explicitly original extension using the SD on p.357. Published survey counts do not verify response order, sampling implementation, or present-day satisfaction.

Complete the practice, then return to the module overview.